

Appendix C to the 2019 Georgia Milestones Growth Model
Report:
Investigation of Potential Ceiling and Floor Effects.

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Abstract

DRAFT REPORT - DO NOT CITE!

1 Introduction

In the 2014-2015 academic year Georgia transitioned from its previous assessment program, the Criterion-Referenced Competency Tests (CRCT), to the [Georgia Milestones Assessment System](#). The transition included numerous changes to the assessment system including the incorporation of new performance standards. As other states have gone through similar assessment transitions, many have observed ceiling and floor effects in the new assessments (i.e. a relatively large proportion of students scoring at/near the scale extremes). These assessment ceilings/floors can telegraph onto the Student Growth Percentile (SGP) calculations.

With perfect data and model fit, the expectation is that the majority of SGPs for students scoring at or near the lowest obtainable scale score (LOSS) will be low (preferably less than 5 and not higher than 20), and that SGPs for students with the highest obtainable scale score (HOSS) will be high (higher than 95 and not less than 80). Ceiling effects in growth measures are somewhat more problematic than floor effects because students that consistently receive the highest scores are given lower than expected growth percentiles and are therefore negatively impacted. Conversely, the consistently lowest achieving students have higher estimated SGPs than expected. This could possibly conceal unacceptably low growth that might otherwise be identified and addressed.

In part, these problems are caused by the way in which a “percentile” is most typically defined to begin with, and the inability of the assessments (and therefore the SGP model) to make granular distinctions between kids who score at the extremes of the test year after year. As an example, if a group of students were given a relatively easy test and 20% of them received a perfect score, these students would be defined as being in the 80th percentile of achievement because they scored higher than 80% of their peers. This is somewhat misleading however, because their score was *equal to* or greater than *all* of their peers and so could be also described as achieving at the 99.9th percentile under other equally valid definitions.

To extend this heuristic from achievement to growth, if 50% of those top students also scored perfectly on the next test, we might estimate that they had 50th percentile growth. Although there is nothing *technically* incorrect about this estimate since their growth is fairly typical for their academic peer group, it is an inadequate or unsatisfactory assessment of their growth because they have consistently attained the highest levels. Furthermore, if it is typical for their peers to maintain perfect scores then even small deviations from a perfect score could produce *low growth* SGP estimates.

Typically only a few students are impacted by ceiling and floor effects, making them difficult to detect using traditional SGP diagnostic tools. The Center for Assessment has recently added “Ceiling/Test Effects” indicators to the SGP model goodness of fit plots and is providing all clients even more rigorous diagnostic and descriptive analyses through this Appendix to the annual technical report. This report includes:

1. Scatter plots of the current and the most recent prior year’s test score distributions to indicate ceilings or floors in the data used in growth calculations.
2. Box plots showing the range and distribution of SGPs for *only* the highest and lowest achieving students in the current year.¹

¹Ranked SIMEX measurement error corrected SGPs are the “official” SGP in Georgia, and are used exclusively in this report.

2 Prior- and Current-Year Score Distributions

The marginal and conditional distributions of test scores can serve as a preliminary indicator of potential ceiling or floor effects in the calculation of student growth percentiles. Some minor problems could occur if these characteristics are present in either the prior- or current-year scores, and are particularly likely when present in both.

The plots in the following sections depict distributions for the current year and the most recent prior year used in the SGP calculations for each content area and grade level. These plots start with a basic scatter plot of each student's scores to show their conditional (joint) score distributions, and each point is depicted as the estimated SGP value based on their scores². On top of this is layered 1) **green contour lines** to provide a better sense of the score distribution density, 2) three **non-linear magenta lines** identifying the bivariate relationship between prior and current scores at the 5th, 50th and 95th percentiles³, 3) **red dotted lines** that represent the cutoff for the highest and lowest 25+ current scale scores (corresponding with the first and last rows of the fit plot table), and 4) **rug plots** that depict the marginal distributions (prior scores shown in blue and current scores in red).

Ceiling or floor effects may be indicated by dark shaded SGP values in the extreme top-right or bottom-left corners of the plots. This suggests that staying at the extremes is common, which may lead to odd growth estimates for these high/low achieving students.

2.1 End-of-Grade Content Areas

We see very few issues in all content areas and grades in the 2018 and 2019 Georgia Milestones EOG data. Where minor ceilings appear in either years' data, the opposite year score distributions for these students are well distributed, lessening the concern for a growth ceiling effect. In past years analyses, some students attaining the HOSS in consecutive years had slightly lower SGPs than 99 (very minor ceiling issues). The Georgia DOE made the decision to manually "correct" those SGPs, assigning a SGP of 99 to all kids scoring the HOSS. That correction was added to the SGP configurations for all 2019 Georgia Milestones analyses so that the SGP estimate is overridden to 99 automatically.

²Note that many SGPs are estimated using more than one prior score, and therefore plots may show SGP results from multiple analyses and/or varying SGPs for identical score combinations.

³Produced using quantile regression similar to, but not the same as, that used in calculation of the SGPs.

Fig. C.1: Conditional distributions of current and prior scale scores: ELA.

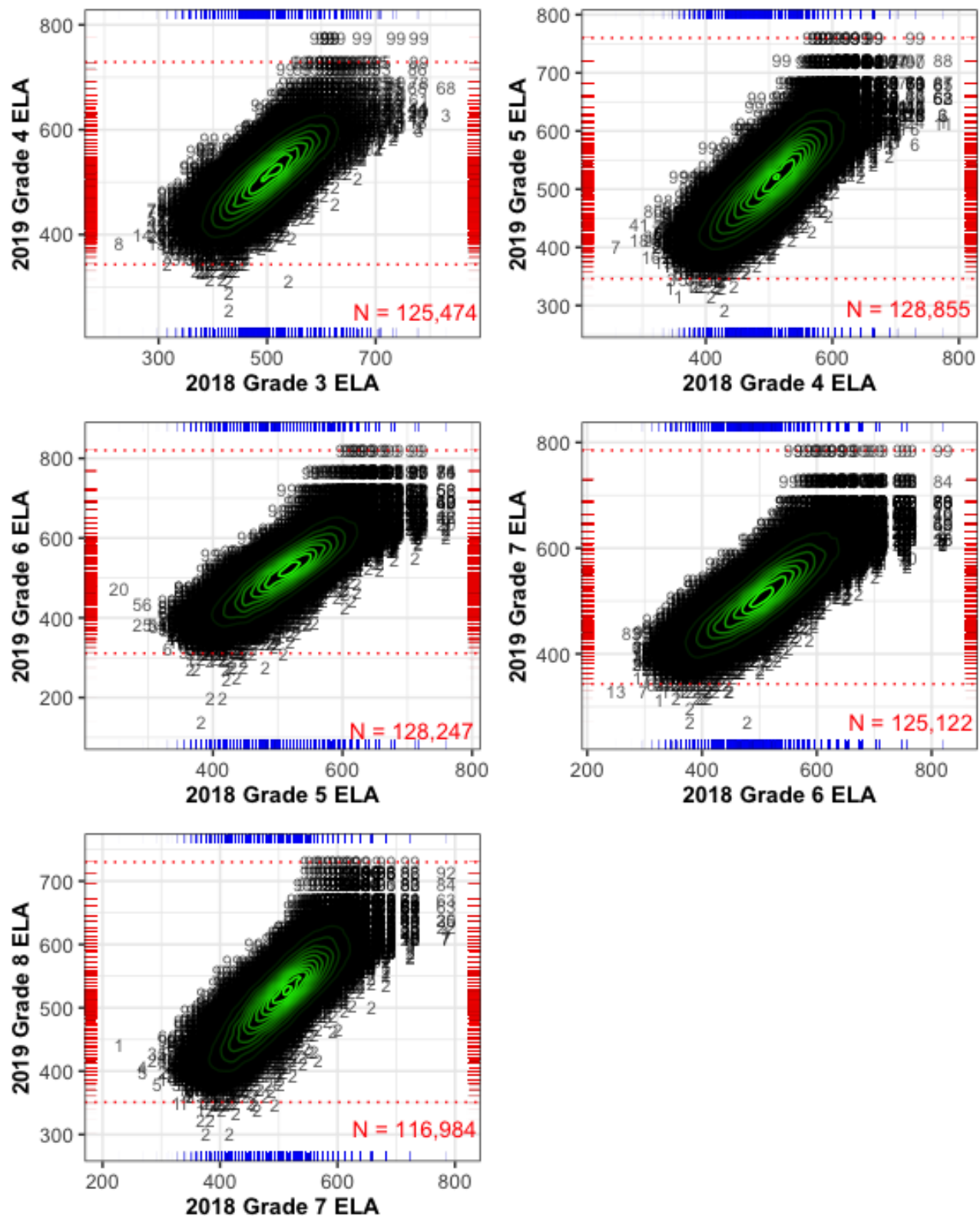
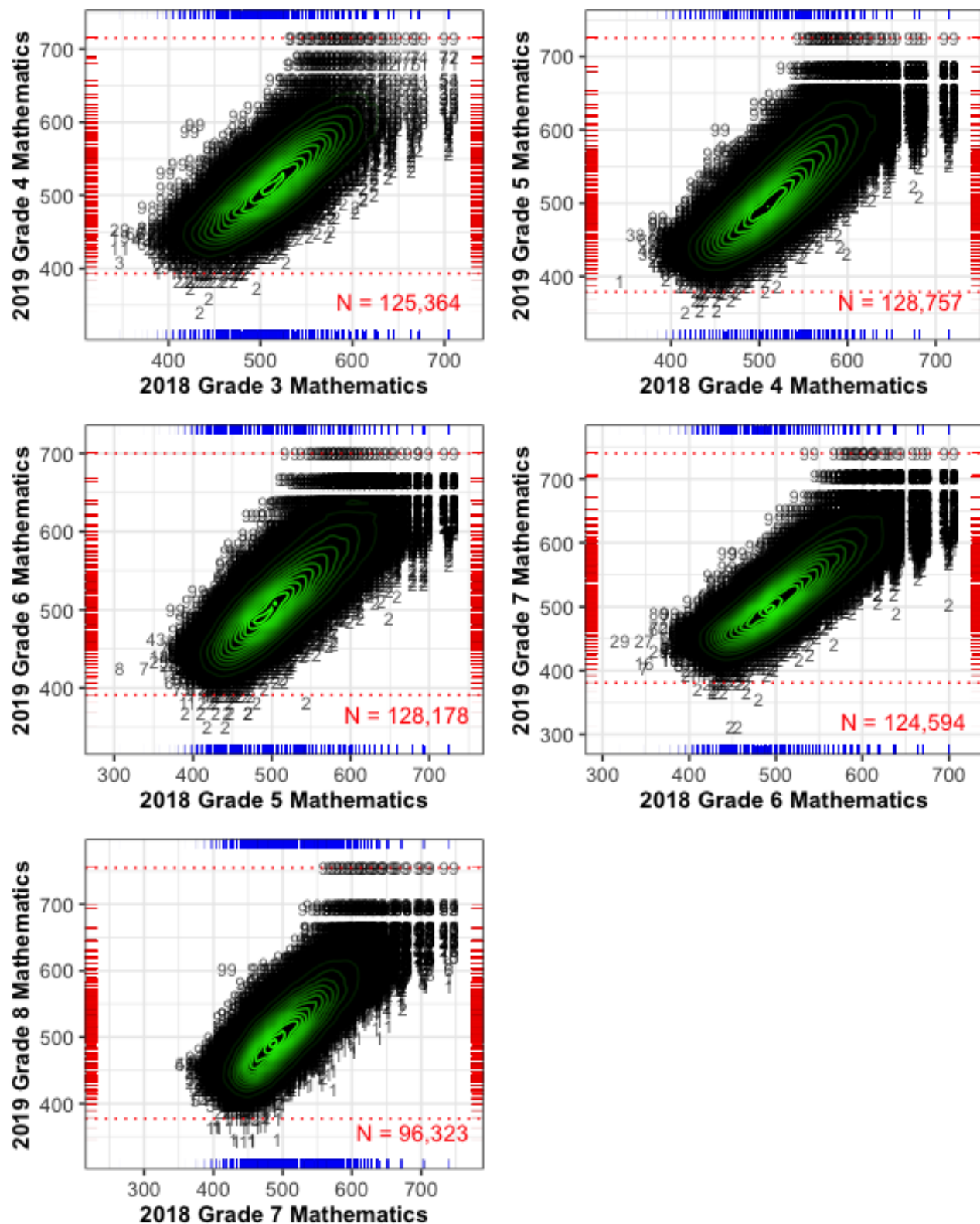


Fig. C.2: Conditional distributions of current and prior scale scores: Mathematics.



2.2 End-of-Course Test Subjects

The conditional density plots for the EOC test subjects are displayed below. The most recent prior is used in each plot to provide insight on the specific academic peer group on which each analysis is based. Some of these norm groups represent atypical student populations (e.g. course repeaters and skipped-year progressions), which have previously shown growth ceiling effects for reasons other than the clustering of scores at the test ceiling. These analyses can have problems due to sparsity of data points in the high achievement regions, which leads the SGP model to struggle to produce adequate and sensible growth estimates. Although these issues are largely corrected manually by overriding the SGP estimates for students attaining the HOSS in the current year (assigning them SGP of 99), some of these atypical norm group analyses present a handful of concerning estimates that require closer examination.

Take, for example, the middle-right plot in Figure C.4 below for American Literature course repeaters which shows that overall the cohort is low achieving (as expected for students retaking the test). The two students in the upper-right hand corner of the plot show how small differences in scores for outliers can have large impacts on SGP estimates. Both students had unusually high scores in 2018 compared to this cohort. One student scored the HOSS (750) on their 2019 assessment and the other had a score just below the HOSS (~700), yet their SGP estimates are 99 and 1 respectively. The student with the HOSS would have been corrected manually to 99 if their model estimate was lower than that, but no correction is made for the other student and they are given an estimate that does not reflect their consistently high achievement.

Another prime example is shown in the lower-left plot of Figure C.5 (a within-year Algebra I repeater analysis) where two students in the upper-right hand corner of the plot are outliers in their cohort but neither have scored the HOSS (785). Again the SGP model produces an unexpectedly low SGP estimate of 5, which is probably an inaccurate assessment of the student's growth since they have consistently attained relatively high and their score did increase slightly.

The handful of students these examples and others show how SGP estimates can fluctuate rather dramatically with very small changes in test scores when the data is sparse at the extremes.⁴

⁴The Georgia DOE has communicated in the past that these outlier students are included in the course repeater cohorts because they failed the course itself despite demonstrating high achievement on the first test.

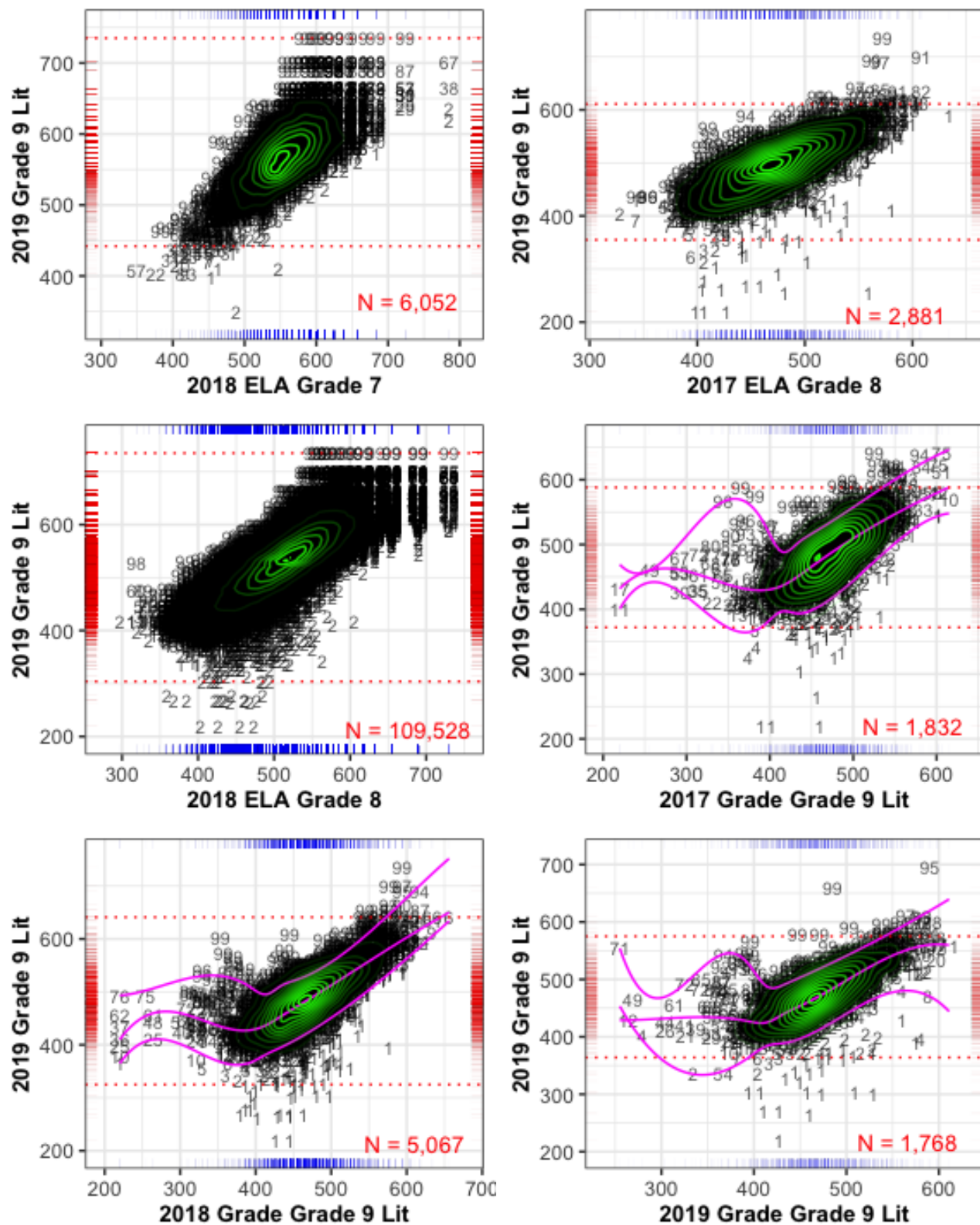
Fig. C.3: Conditional distributions of current and prior scale scores: 9th Grade Lit.

Fig. C.4: Conditional distributions of current and prior scale scores: American Lit.

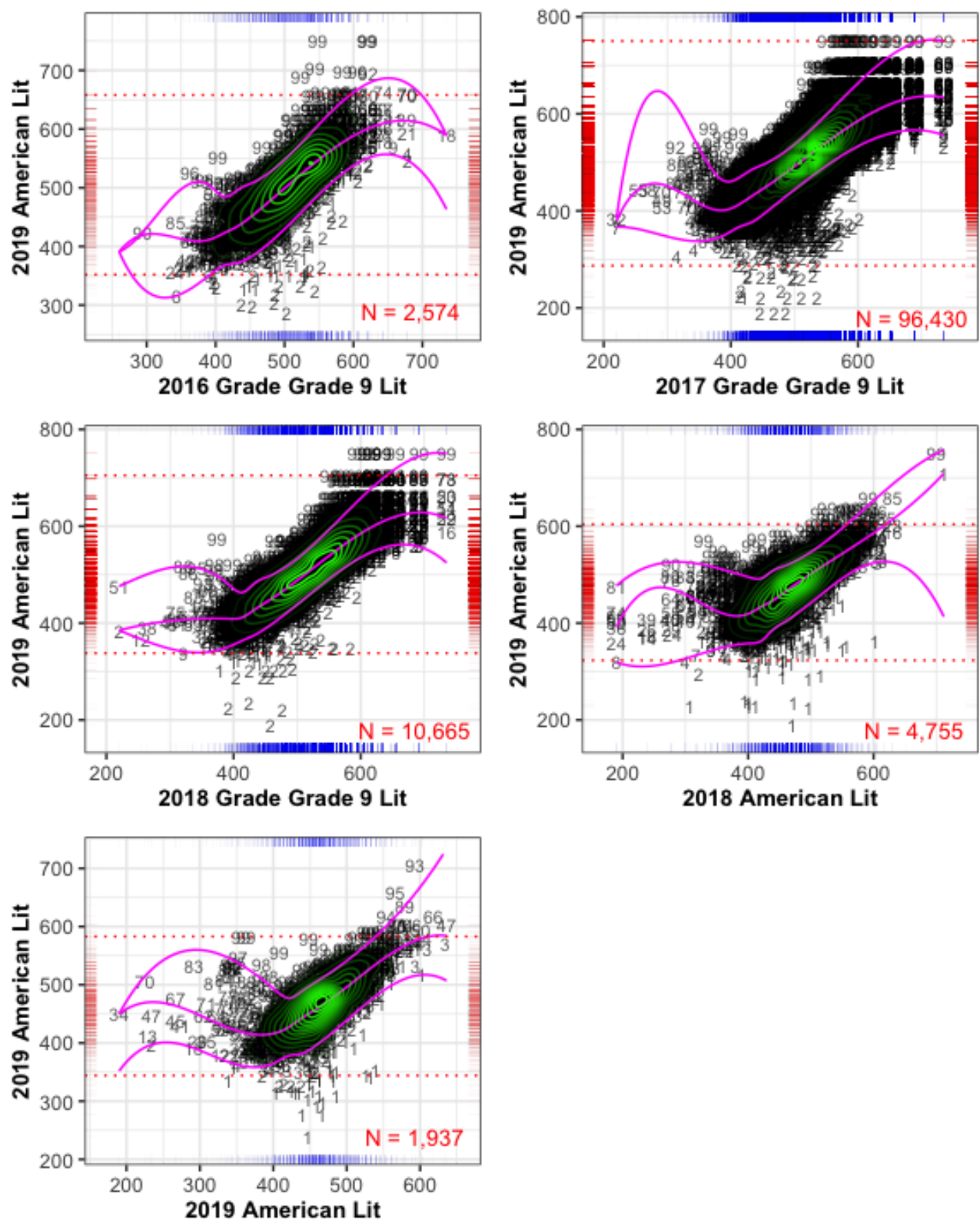


Fig. C.5: Conditional distributions of current and prior scale scores: Algebra I.

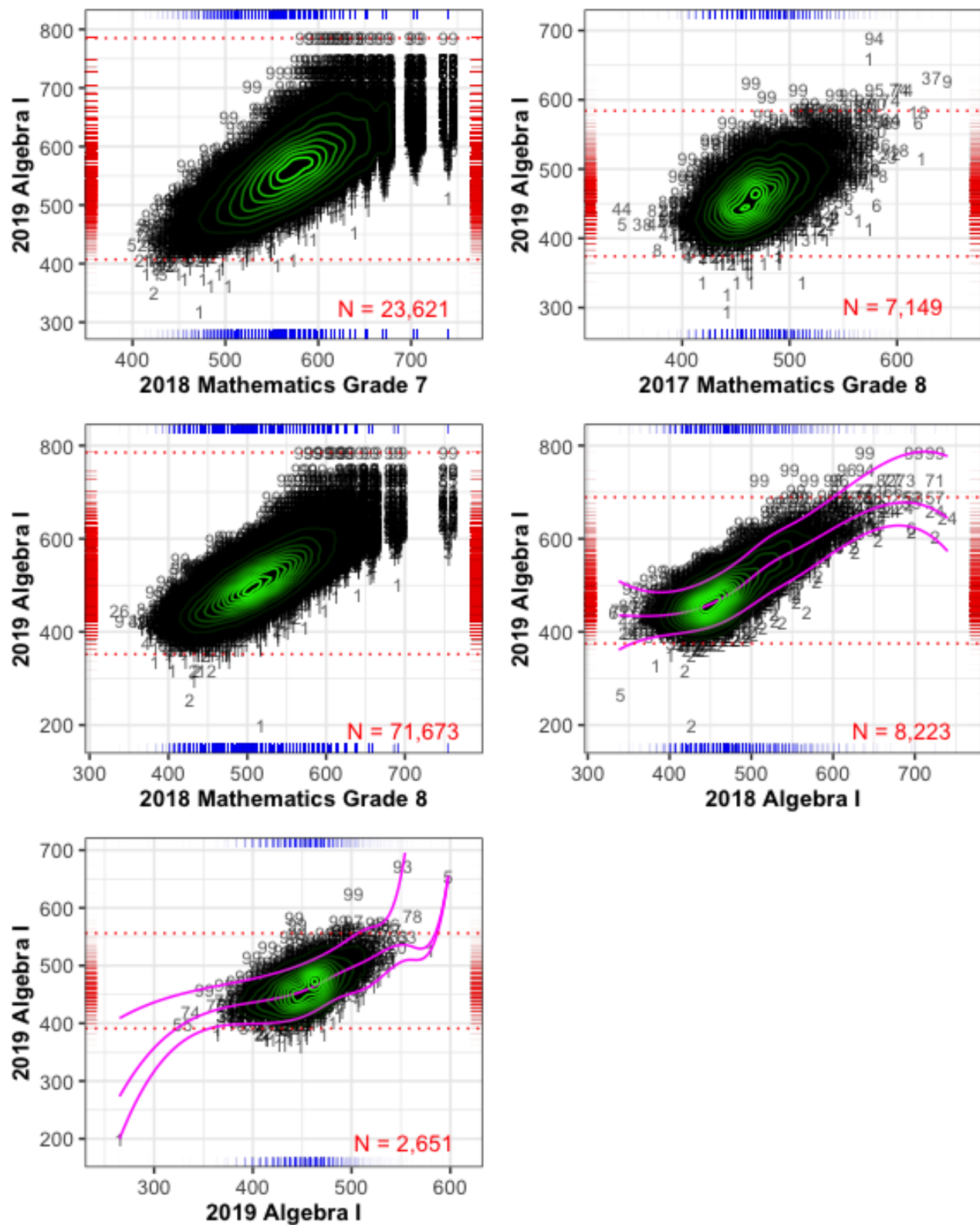


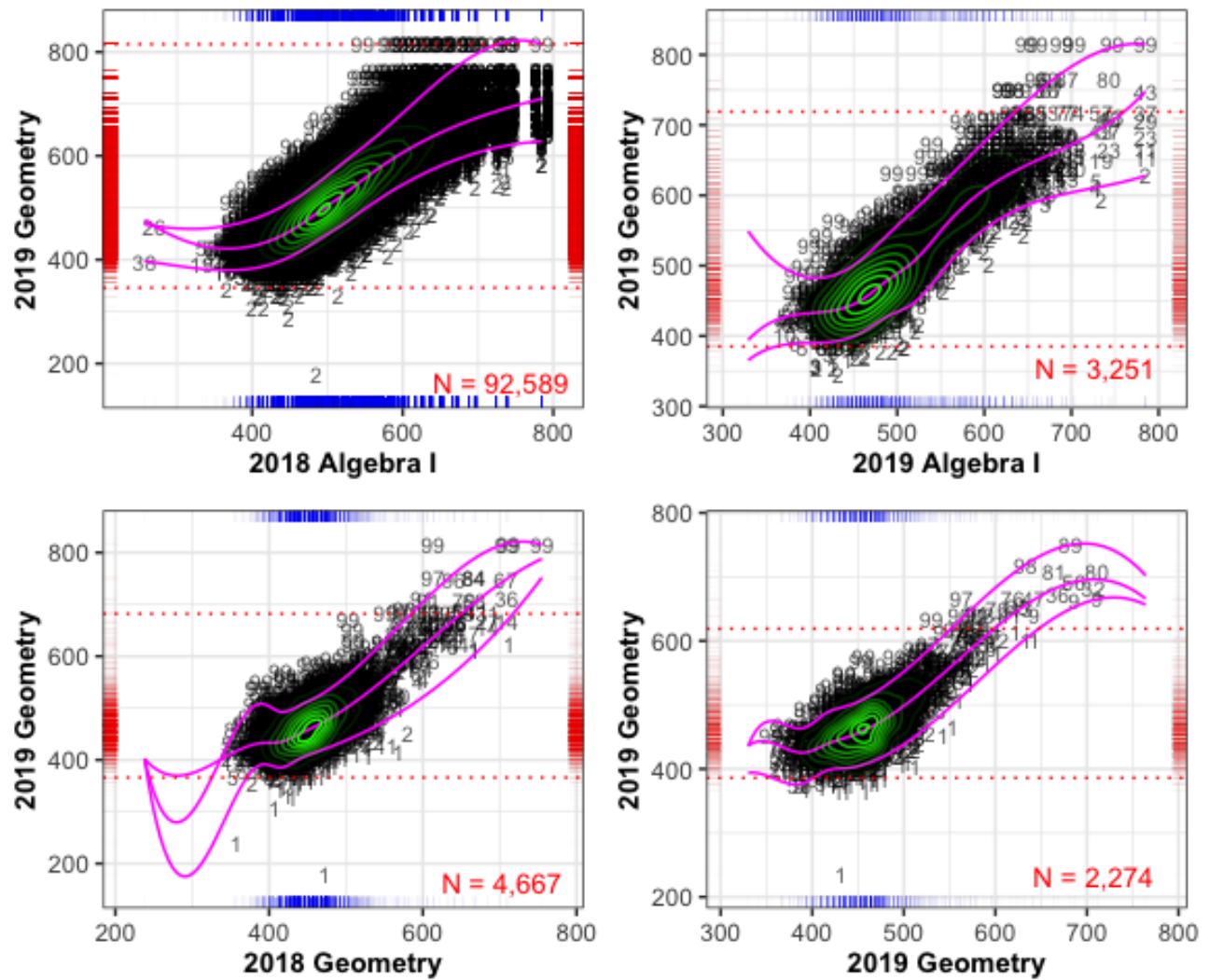
Fig. C.6: Conditional distributions of current and prior scale scores: Geometry.

Fig. C.7: Conditional distributions of current and prior scale scores: Coordinate Algebra.

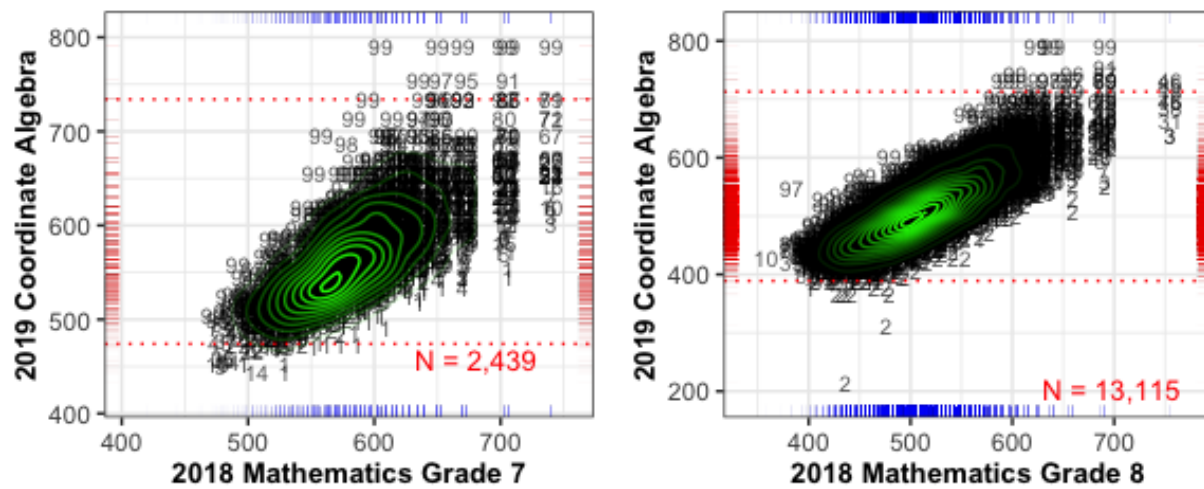
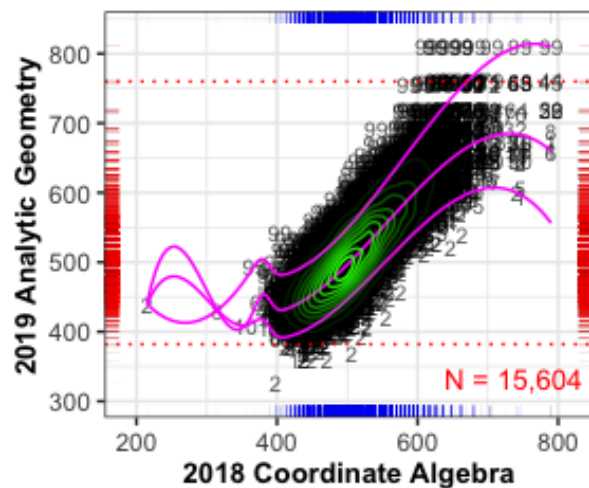


Fig. C.8: Conditional distribution of current and prior scale scores: Analytic Geometry.



3 SGP Ranges for the Highest- and Lowest-Achieving Students

In order to isolate the impact of assessment ceilings/floors on student growth percentile (SGP) calculations, the following section provides box plots of SGP distributions for the highest and lowest achieving students. We are specifically interested in the growth percentiles for students scoring at the highest/lowest obtainable scale score (HOSS/LOSS - i.e. the test ceiling/floor) on the current year test. However, in order to assure that an adequate number of students are included, the first set of plots uses, at a *minimum*, the highest/lowest 25 scores. These plots are provided as a starting point since this roughly corresponds to the number of students in the top and bottom rows of the table included in the SGP model “Goodness of Fit” plots. All students with a score in these students’ range of scores are included. Consequently, the number of students in each box plot may be greater than 25 (the exact number is shown at the margins in red text).

The second set of box plots isolate *only* those students scoring the HOSS/LOSS. These plots may then incorporate a varying number of students depending on the prevalence of a ceiling/floor in the current year. The Ranked SIMEX measurement error corrected SGP is Georgia’s official growth metric, and these values are used exclusively in this report.

The box plots provide several descriptive statistics. The dark line within the box marks the *median* SGP, while the ends (“hinges”) of the boxes correspond to the first and third quartiles (the 25th and 75th percentiles). The upper whisker extends from the hinge to the highest value that is within $1.5 \times \text{IQR}$ of the hinge, where IQR is the inter-quartile range, or distance between the first and third quartiles. The lower whisker extends from the hinge to the lowest value within $1.5 \times \text{IQR}$ of the hinge. Data beyond the end of the whiskers are outliers and plotted as individual points. Evidence of a *lack* of either a ceiling or floor effect would be to have all high achieving students with SGPs near 99 and all low achieving students with SGPs near 1. That is, the desired visual evidence is a solid line at $\text{SGP} = 99/1$.

3.1 EOG Content Areas

The scatter plots in the previous section showed that it is far more typical for students to score the HOSS on the Mathematics assessment than in ELA. In previous years this translated to moderate ceiling effect in Mathematics. As of 2017 the Georgia Department of Education has decided to award students scoring the HOSS an SGP of 99 regardless of their SGP model estimate to address these issues, and beginning this year this correction has been incorporated into the SGP calculation itself so that all results in the following boxplots appear perfect as expected.

Fig. C.9: EOG SGP distributions for highest and lowest 25+ scale scores by content area and grade level.

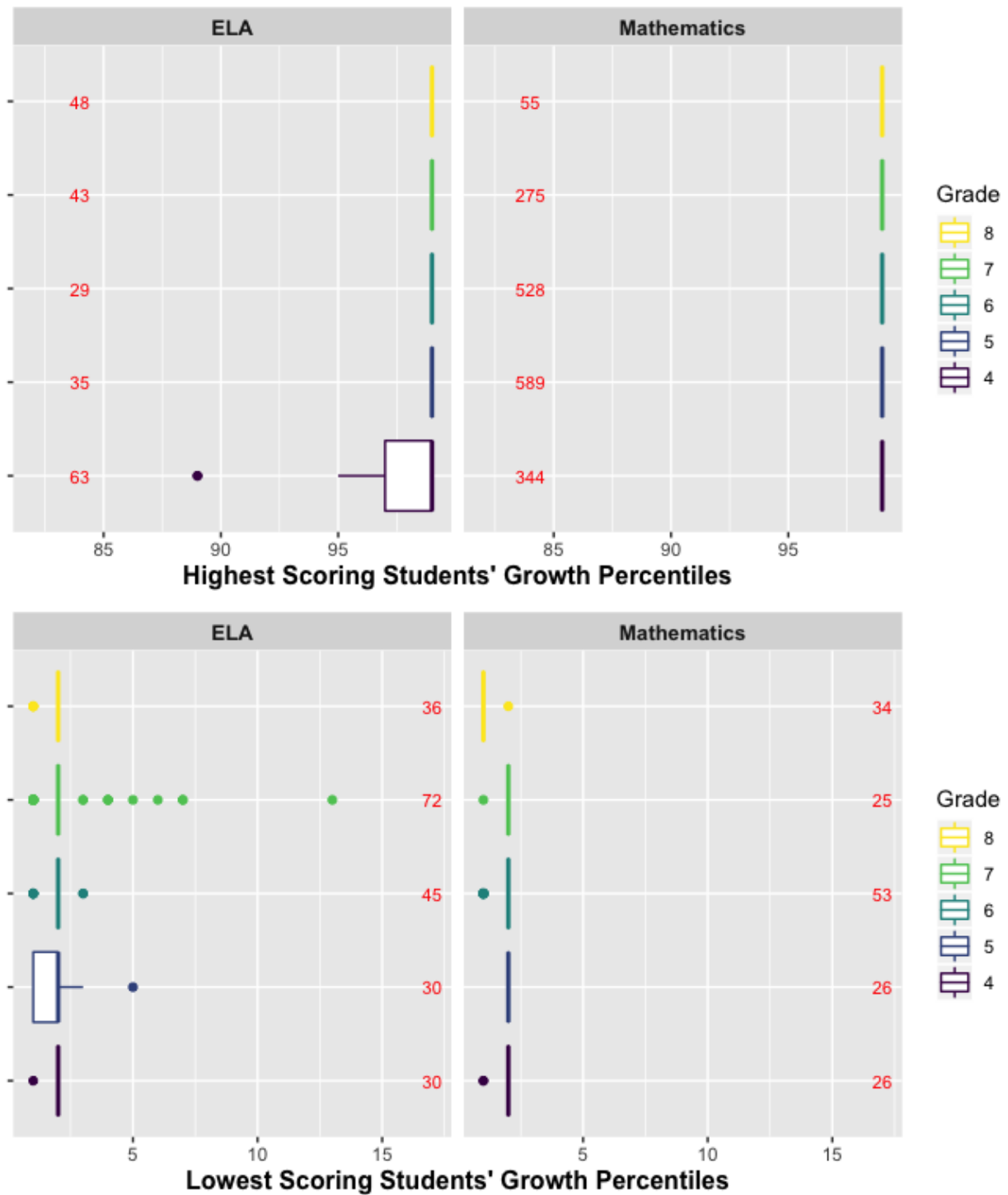
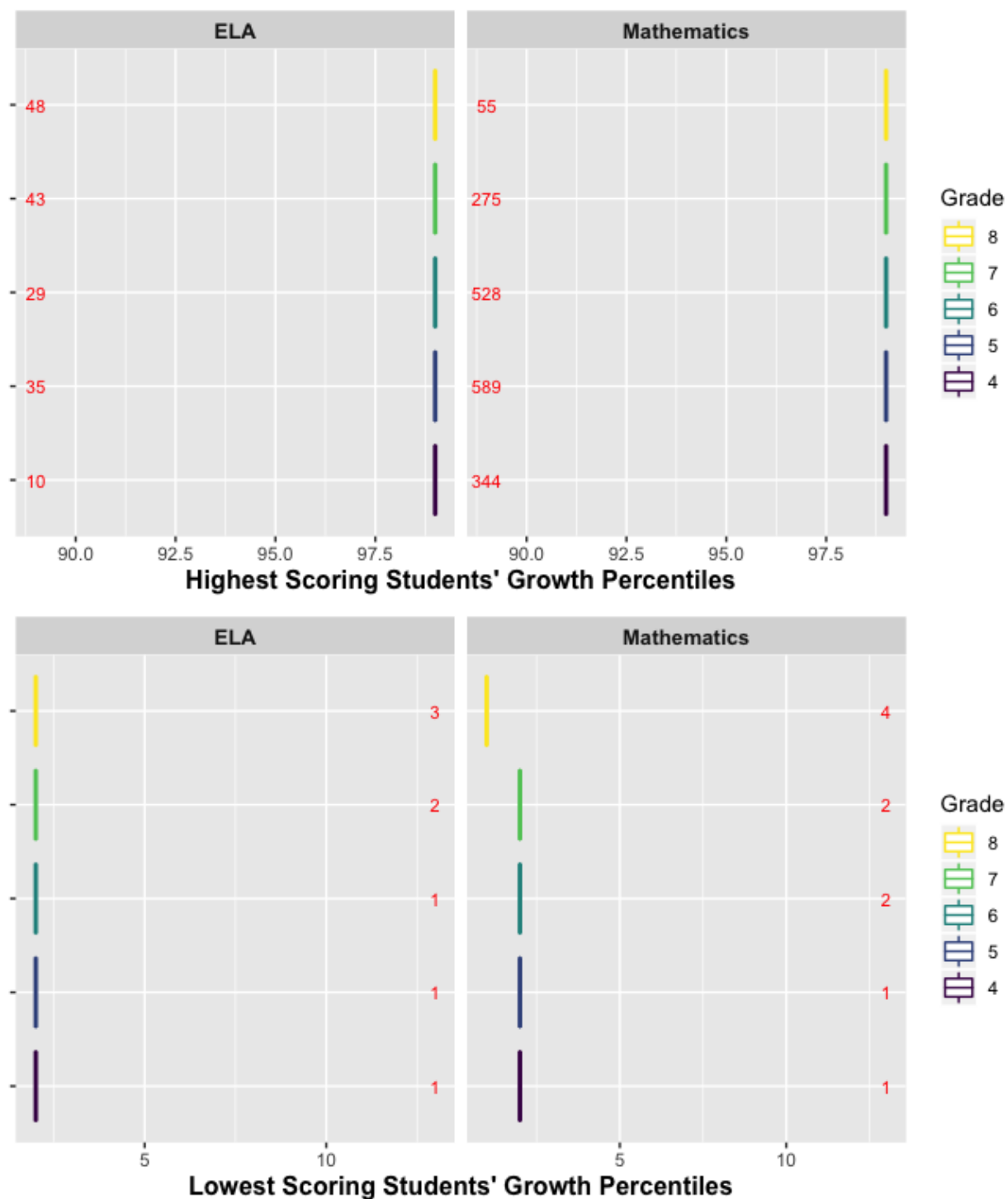


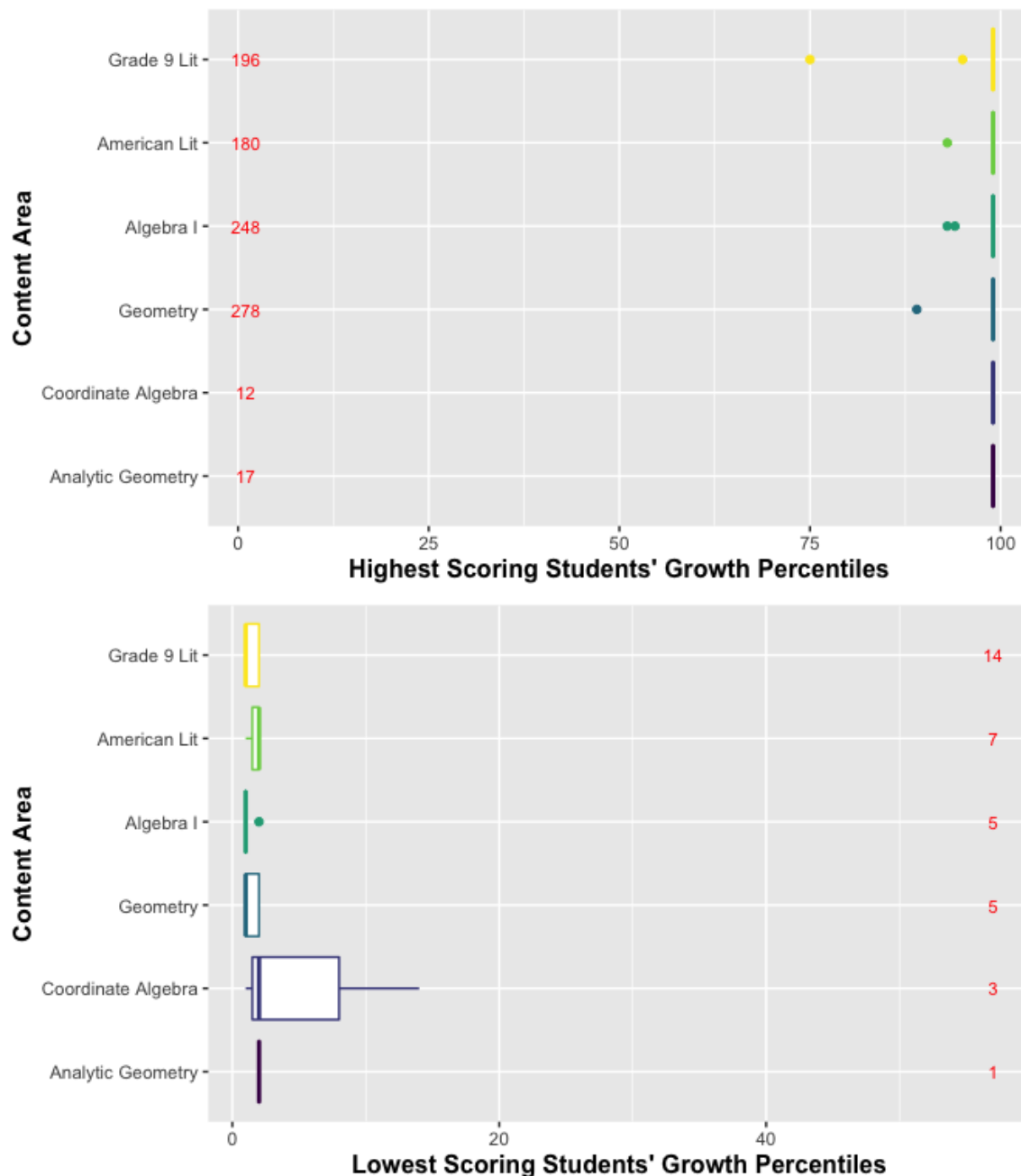
Fig. C.10: EOG SGP distributions for the HOSS and LOSS scores by content area and grade level.



3.2 EOC Subjects

The end-of-course subject results are shown here *only for students scoring exactly the HOSS and LOSS* respectively. There are several subjects for which small ceiling effects are evident. However, as mentioned in the previous section, these relatively low SGPs occur in atypical cohort analyses and are associated with students scoring below the HOSS (they have the highest/lowest *observed* scale scores) and so are not automatically corrected to 99. In order to adequately address any concerns all EOC subjects are also shown in separate plots where the results are disaggregated by the most recent prior test used in each analyses.

Fig. C.11: EOC SGP distributions for the HOSS and LOSS scores by content area.



The following box plots disaggregate each EOC subject by the most recent prior to reflect their constituent norm groups more closely. These plots suggest that only a handful of students in the course repeater cohorts have potentially problematic SGP estimates.

The plots for the Grade 9 Literature skipped-year course-repeater progression is notable. This progression can also be seen in Figure C.3 above. Here the student who scored much higher than the rest of the cohort in both test administrations (although higher on the first test) has an unexpectedly low SGP of 75. It is not surprising that the SGP estimates of growth are questionable for outliers in the data such as this.

Fig. C.12: EOC SGP distributions for the HOSS and LOSS scores by norm group: 9th Grade Literature.

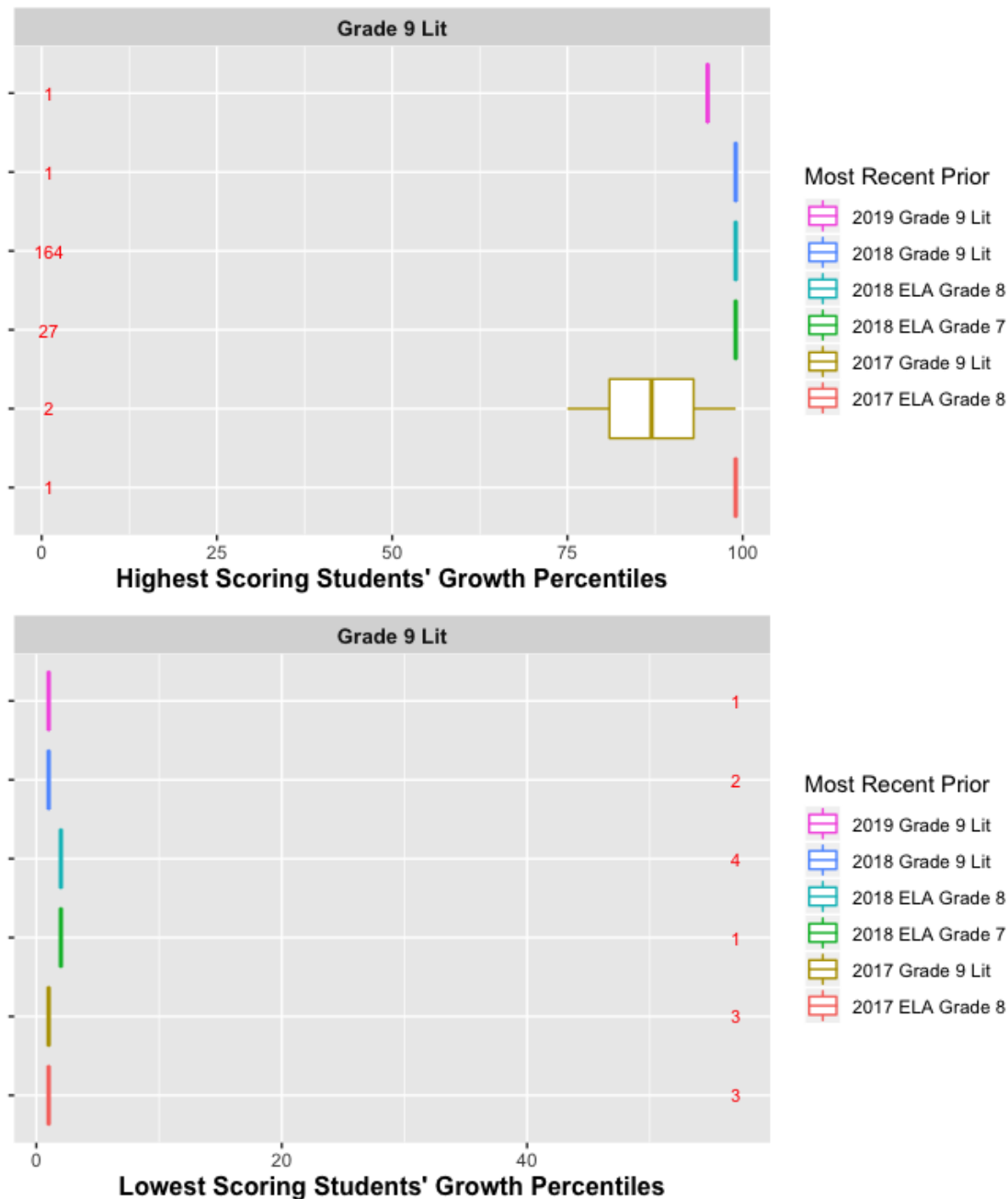


Fig. C.13: EOC SGP distributions for the HOSS and LOSS scores by norm group: American Literature.

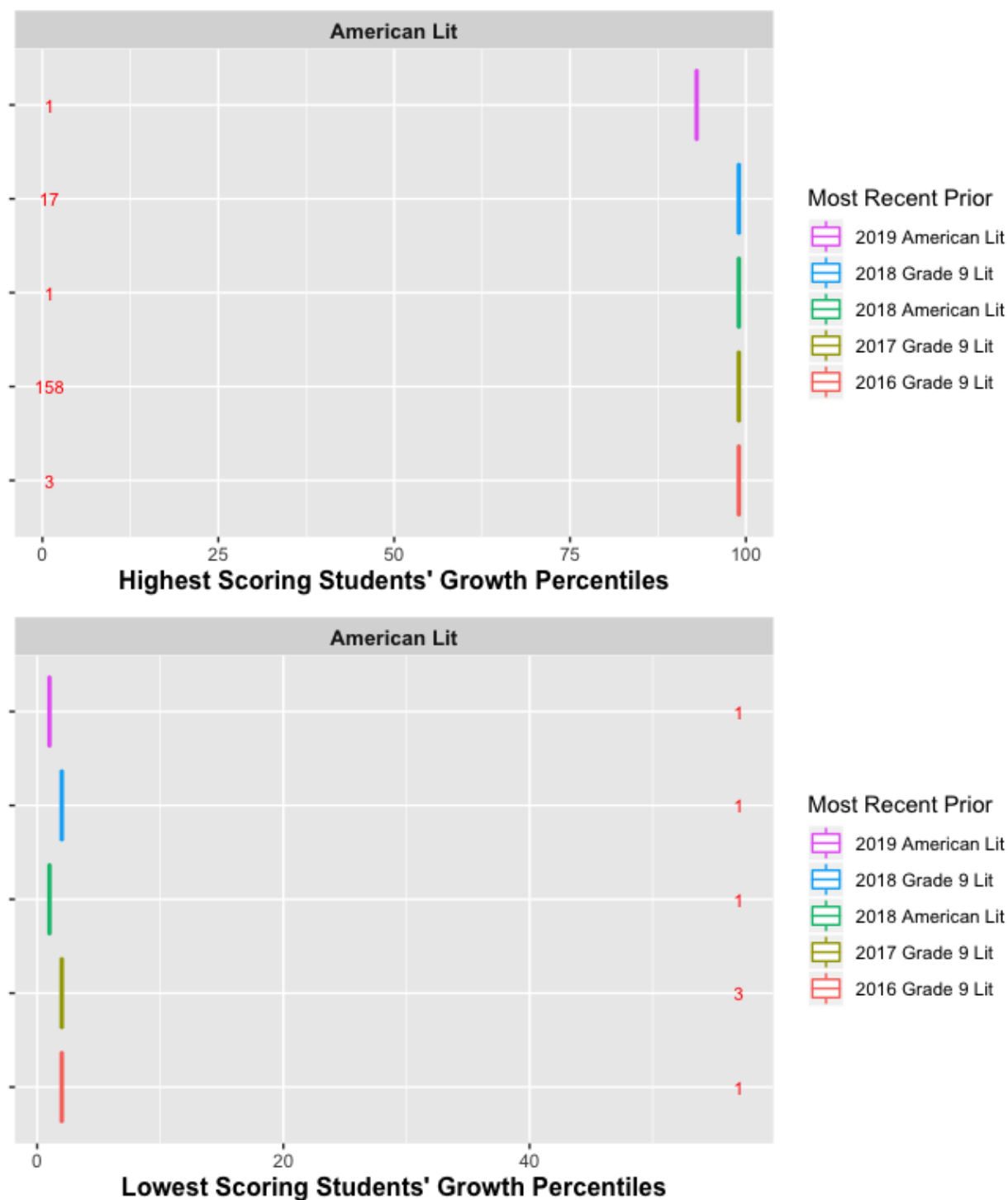


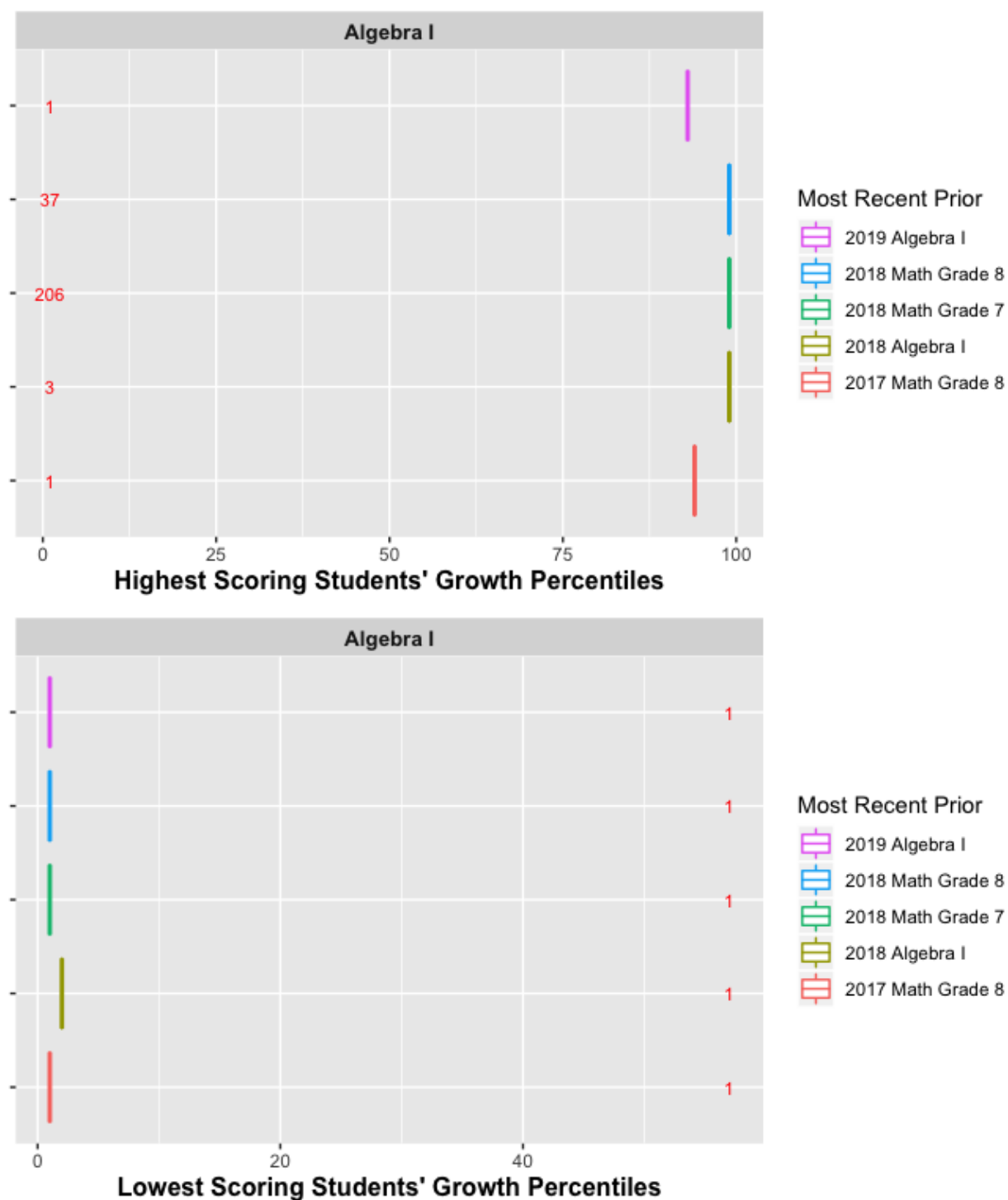
Fig. C.14: EOC SGP distributions for HOSS and LOSS scores by norm group: Algebra I.

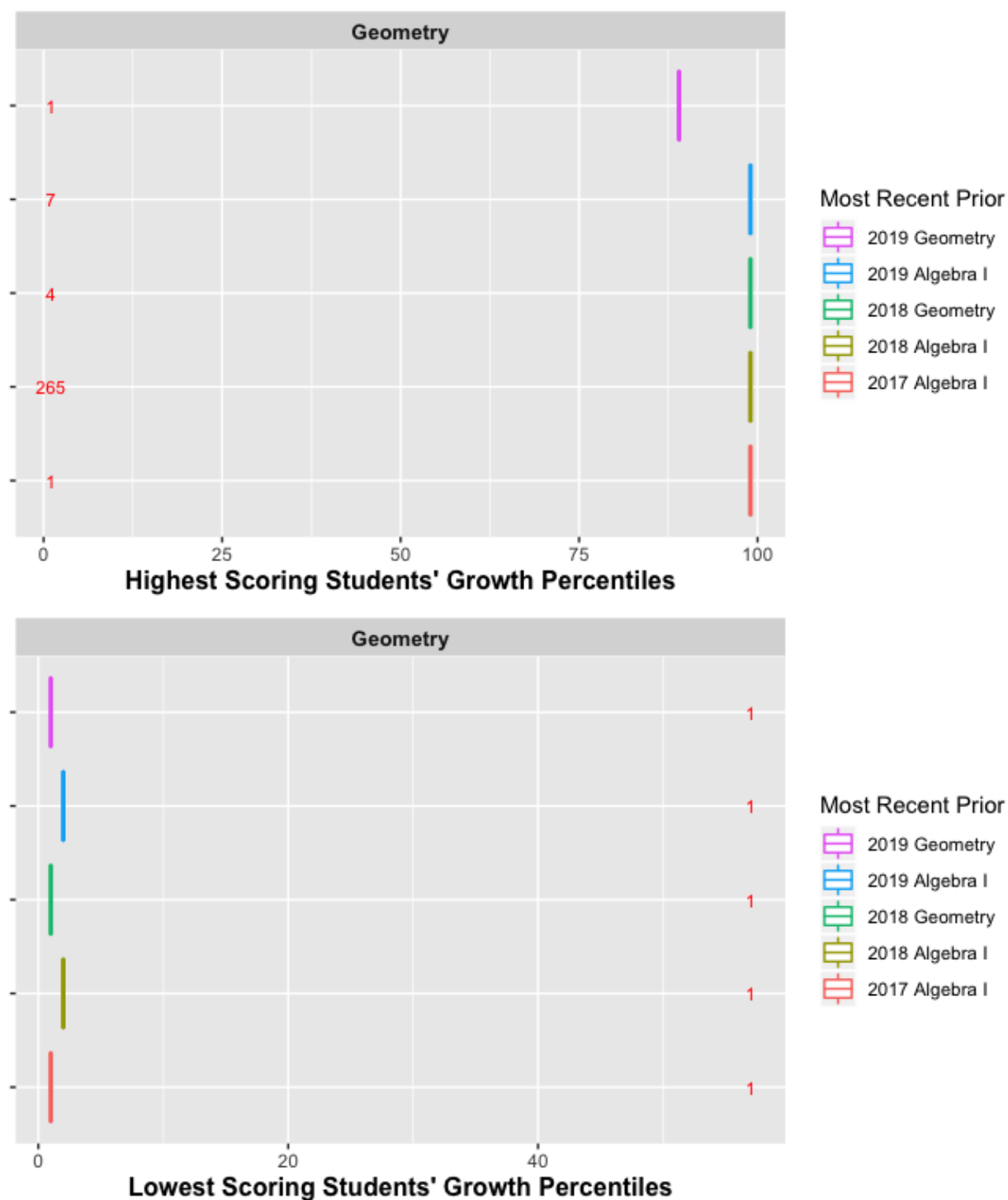
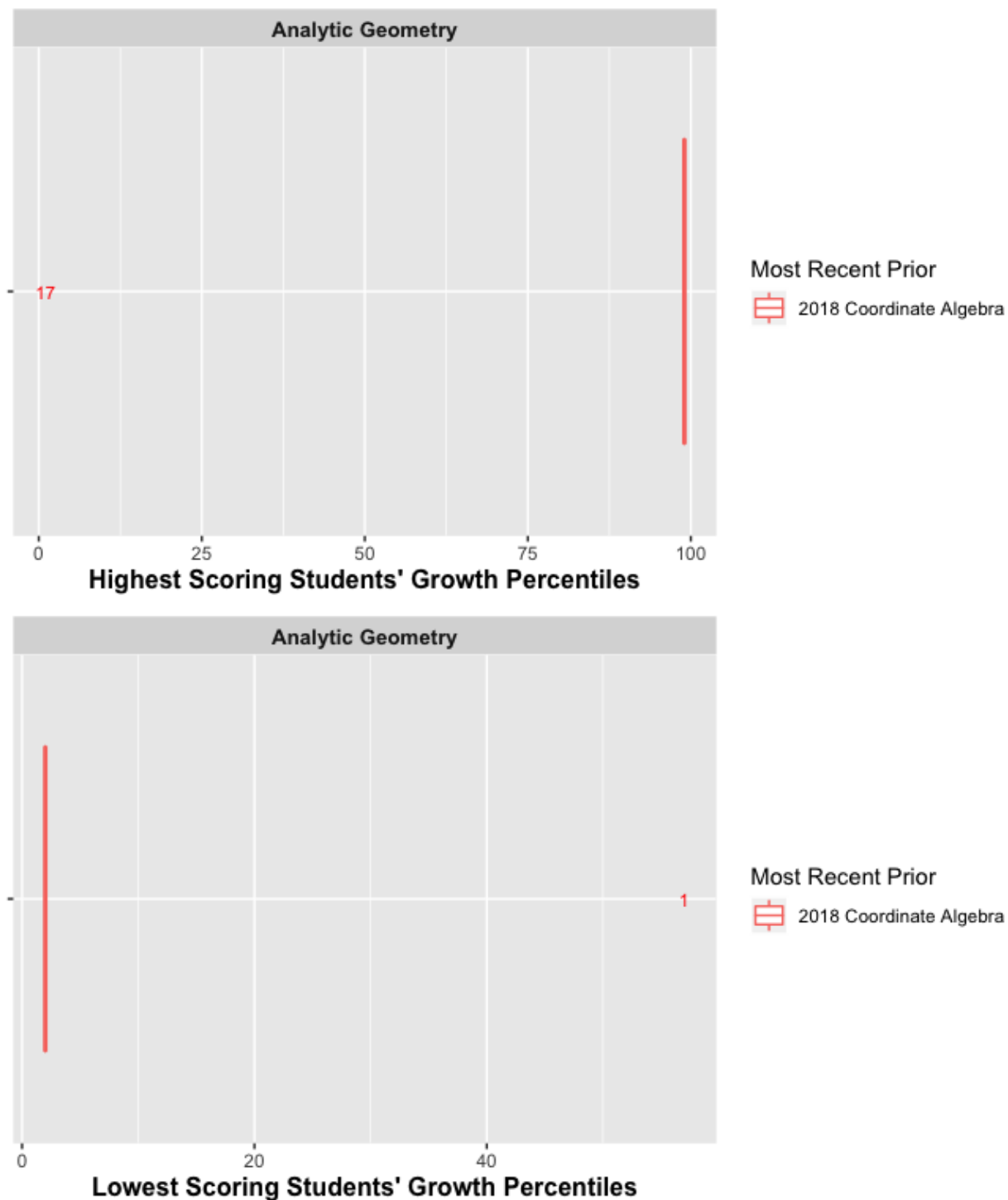
Fig. C.15: EOC SGP distributions for the HOSS and LOSS scores by norm group: Geometry.

Fig. C.16: EOC SGP distributions for the HOSS and LOSS scores by norm group: Coordinate Algebra.



Fig. C.17: EOC SGP distributions for the HOSS and LOSS scores by norm group: Analytic Geometry.



4 Discussion

Overall there is little evidence of floor or ceiling effects in the 2019 Georgia Milestones SGP analyses. Some scores for EOC test repeater progressions suggest that minor problems may exist for a small number of students. When ceiling or floor effects are encountered, there are several ways in which they can be “corrected” manually or analytically. These include (but are not limited to):

1. Convert all students scoring at the HOSS (LOSS) to 99 (1). Georgia has used this manual correction beginning in 2017.
2. Run SGP analyses with more granular scores. For example, many tests that use Item Response Theory (IRT) to analyse test results provide scaled scores that enforce an artificial ceiling (floor), but also have more granular achievement scores available (IRT θ estimates).
3. Leave the results without a correction.